SeedHammer II — the reasonably complex wallet

A four-tier degrading vault, taken end to end in **both** its wrappings: taproot and wsh. Six seeds in; two engraved backups out; the device agreeing with the host on every address, by two different routes.

> Every command in this document ran, and its real stdout, stderr and exit code are reproduced. Every device screen is a frame the emulator rendered while an assertion held. Regenerate with `bash transcript_rcw.sh > transcript_rcw.txt` then `python3 build_pdf_rcw.py`; the builder refuses a transcript produced by a different version of its generator.

> **Test material.**
> The six seeds come from entropy 0x000…001 through 0x000…006 and the three passphrases are printed in the generator. Never put funds behind any of it.

What the starting representation actually is

```
The starting representation of "our reasonably complex wallet" is a BIP-388
WALLET POLICY TEMPLATE: a miniscript expression in which every key is a
PLACEHOLDER -- @0 through @5 -- followed by that slot's derivation path and a
<0;1> multipath suffix for the receive/change chains.

It is NOT a descriptor, NOT an address, and NOT a set of keys. It is the SHAPE
of the wallet with key-shaped holes in it, and everything below is produced
from it plus the six seeds. That distinction is the whole reason a keyless
template can be engraved separately from the keys that fill it.

The four tiers, in the order the policy states them:

  tier 1   @0 and @1 and @2 and sha256(H1)          -- at any time
  tier 2   @3 and @4 and sha256(H2)                 -- after older(32768), relative
  tier 3   @5 alone                                 -- after height 1173520, absolute
  tier 4   sha256(H3) alone                         -- after height 1383520, absolute

Tier 4 has NO KEY. Whoever learns H3's preimage can spend it alone once that
height passes, which is why every encode below carries --experimental and why
the preimages are as sensitive as the seeds.

Two wrappings, which are DIFFERENT WALLETS with different addresses:

  tr   taproot, NUMS internal key so there is no key-path spend; the four tiers
       are four separate taptree leaves. Keys at m/270028'/0'/8'/0'.
  wsh  segwit v0; the taptree FLATTENS into one script, so the tiers become an
       or_i chain and multi_a becomes multi. Keys at m/270028'/0'/9'/1'.
```

Where the policy came from — and why the table above is now gated

The four-tier table on the previous page reads like the *source* of this wallet. It is not. The arrow runs the other way, and the provenance is short enough to print in full:

```
The four-tier table above reads like the source of this wallet. It is not. The
arrow runs the other way, and the provenance is short enough to print:

$ git -C /scratch/code/shibboleth/mnemonic-engrave log --oneline --diff-filter=A -- design/journeys/inputs-hashvault/wallet-
policy-hashvault.txt
f39958b journeys: the hashlock vault — six seeds, three passphrases, four degrading tiers
[exit 0]

$ git -C /scratch/code/shibboleth/mnemonic-engrave log --oneline --diff-filter=A -- design/fixtures/reasonably-complex-walle
t/tr.policy
5b15b18 fixtures: name "our reasonably complex wallet", and measure both wrappings
[exit 0]

And this journey's copy is that fixture with one path substitution:
$ grep -n policy-tr.txt /scratch/code/shibboleth/mnemonic-engrave/design/journeys/derive-rcw-keys.sh
133:sed "s#270028'/0'/0'/0'#270028'/0'/8'/0'#g" "$FIX/tr.policy"  > "$IN/policy-tr.txt"
[exit 0]


So: the operator's English spec was turned into a policy string BY HAND, once,
in the hashlock-vault cycle. Everything since is copy plus `sed`. The prose
table is a DESCRIPTION written afterwards, and until now nothing tied the two
together — it was accurate because it was checked once by hand, which is the
state every other unverified claim in this repo was in before it turned out to
be wrong.

check_tiers.py closes that: it parses the real policy and asserts each tier,
per wrapper, including that EXACTLY ONE tier is keyless.

$ python3 /scratch/code/shibboleth/mnemonic-engrave/design/journeys/check_tiers.py /scratch/code/shibboleth/mnemonic-engrav
e/design/journeys/inputs-rcw/policy-tr.txt tr
  tr tier 1: OK   @0+@1+@2 and sha256(H1), NO timelock
  tr tier 2: OK   @3+@4 and sha256(H2) after older(32768) [relative]
… clipped; full text in transcript_rcw.txt
```

> The operator's English spec was turned into a policy string
> **by hand, once**
> , in the hashlock-vault cycle. Everything since is a byte-for-byte copy plus one `sed` path substitution. The prose table is a **description written afterwards** — and until this run, nothing tied the two together. It was accurate because it had been checked once by hand, which is precisely the state every claim in this repo occupied right before it turned out to be wrong.

The gate, and proof it can fail

`check_tiers.py` parses the real policy string and asserts each tier against it, per wrapper — including that **exactly one** tier is keyless, so neither losing that property nor gaining a second keyless branch passes quietly.

```
. Give tier 4 a key -- the one change that would make
this wallet importable, and would silently make the table above false:

$ python3 /scratch/code/shibboleth/mnemonic-engrave/design/journeys/check_tiers.py /scratch/code/shibboleth/mnemonic-engrav
e/design/journeys/out/rcw/policy-tr-KEYED-TIER4.txt tr
  tr tier 1: OK   @0+@1+@2 and sha256(H1), NO timelock
  tr tier 2: OK   @3+@4 and sha256(H2) after older(32768) [relative]
  tr tier 3: OK   @5 alone after after(1173520) [absolute]
  tr tier 4: FALSE sha256(H3) ALONE after after(1383520) — NO KEY
      branch as written: and_v(v:after(1383520),and_v(v:pk(@5/270028'/0'/8'/0'/<0;1>/*),sha256(4743d7c47df21d29e3ed3dfec5d0c
0a884ccc270
  tr: 0 keyless branches, expected exactly 1
FAIL[tr]: the transcript's tier table does not match the policy: tier 4, keyless-count
[exit 1]
```

The negative control gives tier 4 a key — the single change that would make this wallet importable by third-party software, and that would silently falsify the table. The gate catches it on two independent grounds: tier 4's own claim, and the exactly-one-keyless count.

Written after the first version of this checker reported a false FAIL on tier 2. Tier 2 is a *nested* `and_v` — `older` wrapping `sha256` and `multi_a` — and a single-level regex grabs the inner group, so the branch looks like it has lost its timelock. Structure needs bracket balancing, not patterns. The fix is in the function's own doc comment so the next person does not repeat it.

The two policies

Same four tiers, two wrappings, and they are **different wallets**. The wsh form is not the tr form with a different prefix: `multi_a` becomes `multi` because CHECKSIGADD is tapscript-only, and the taptree — four separate leaves — flattens into one script, so the tiers become an `or_i` chain.

> The consequence is worth stating: in `tr` a spend reveals only the leaf used. In `wsh` **every** spend reveals all four tiers, including the other two hashlock digests and both deadlines.

```
The two policies, as generated by derive-rcw-keys.sh:

$ cat /scratch/code/shibboleth/mnemonic-engrave/design/journeys/inputs-rcw/policy-tr.txt
tr(50929b74c1a04954b78b4b6035e97a5e078a5a0f28ec96d547bfee9ace803ac0,{and_v(v:sha256(ede0b28a805feca09a77c48f82babc1249c79aa8
40de94fa4a85bf55a453c813),multi_a(3,@0/270028'/0'/8'/0'/<0;1>/*,@1/270028'/0'/8'/0'/<0;1>/*,@2/270028'/0'/8'/0'/<0;1>/*)),{a
nd_v(v:older(32768),and_v(v:sha256(9ce09a8df1d1f8bc8892441a9eb30d0a18bc7db81bb0fb3f54c6d9064e7b76ad),multi_a(2,@3/270028'/
0'/8'/0'/<0;1>/*,@4/270028'/0'/8'/0'/<0;1>/*))),{and_v(v:after(1173520),pk(@5/270028'/0'/8'/0'/<0;1>/*)),and_v(v:after(13835
20),sha256(4743d7c47df21d29e3ed3dfec5d0c0a884ccc2708637dddf771c36d214056954))}}})
[exit 0]

$ cat /scratch/code/shibboleth/mnemonic-engrave/design/journeys/inputs-rcw/policy-wsh.txt
wsh(or_i(and_v(v:sha256(ede0b28a805feca09a77c48f82babc1249c79aa840de94fa4a85bf55a453c813),multi(3,@0/270028'/0'/9'/1'/<0;1>/
*,@1/270028'/0'/9'/1'/<0;1>/*,@2/270028'/0'/9'/1'/<0;1>/*)),or_i(and_v(v:older(32768),and_v(v:sha256(9ce09a8df1d1f8bc8892441
a9eb30d0a18bc7db81bb0fb3f54c6d9064e7b76ad),multi(2,@3/270028'/0'/9'/1'/<0;1>/*,@4/270028'/0'/9'/1'/<0;1>/*))),or_i(and_v(v:a
fter(1173520),pk(@5/270028'/0'/9'/1'/<0;1>/*)),and_v(v:after(1383520),sha256(4743d7c47df21d29e3ed3dfec5d0c0a884ccc2708637ddd
f771c36d214056954))))))
[exit 0]
```

Six seeds, six fingerprints

```
key-0: abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon aband
on abandon abandon abandon abandon abandon abandon abandon abandon diesel
key-1: abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon aband
on abandon abandon abandon abandon abandon abandon abandon abandon false
key-2: abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon aband
on abandon abandon abandon abandon abandon abandon abandon abandon kite
key-3: abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon aband
on abandon abandon abandon abandon abandon abandon abandon abandon organ
key-4: abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon aband
on abandon abandon abandon abandon abandon abandon abandon abandon ready
key-5: abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon aband
on abandon abandon abandon abandon abandon abandon abandon abandon surface

TEST MATERIAL: entropy 0x000..001 through 0x000..006. Never put funds behind these.
```

```
  key   fingerprint
  @0    39ec1b6e
  @1    d02bcedf
  @2    5fd78eb3
  @3    92c4c8b2
  @4    086c047e
  @5    cef8224c

6 distinct fingerprints across 6 seeds.
```

The master fingerprint is a property of the
seed
, not of a derivation path, so it is identical in both wrappings. The generator asserts that rather than assuming it — if the two runs ever produced different fingerprints for one seed, something would be very wrong and silence would be the worst outcome.

The keys, and why 8 and 9

```
bg002h is m/270028'/coin'/account'/script', with the level-4 script value ruled
0' = tr and 1' = wsh. The operator selected ACCOUNT 8 for the tr form and
ACCOUNT 9 for the wsh form, so the two wallets hold disjoint keys derived from
the same six masters.

-- tr --
$ head -3 /scratch/code/shibboleth/mnemonic-engrave/design/journeys/inputs-rcw/keys-tr/key-0.xpub
# @0 — tier 1 — 3-of-3 with hash A, spendable at any time
# tr wrapper; seed entropy 0x0000000000000000000000000000000000000000000000000000000000000001 (TEST ONLY)
# origin [39ec1b6e/270028'/0'/8'/0']
[exit 0]

-- wsh --
$ head -3 /scratch/code/shibboleth/mnemonic-engrave/design/journeys/inputs-rcw/keys-wsh/key-0.xpub
# @0 — tier 1 — 3-of-3 with hash A, spendable at any time
# wsh wrapper; seed entropy 0x0000000000000000000000000000000000000000000000000000000000000001 (TEST ONLY)
# origin [39ec1b6e/270028'/0'/9'/1']
[exit 0]

############################################################
########## tr
############################################################
```

tr — the cards

```
Fingerprints are declared (F-227): all six slots share one origin path,
so without them a gathered key card matches every slot and seating must
refuse the whole set.

$ /scratch/code/shibboleth/descriptor-mnemonic/target/debug/md encode tr(50929b74c1a04954b78b4b6035e97a5e078a5a0f28ec96d547b
fee9ace803ac0,{and_v(v:sha256(ede0b28a805feca09a77c48f82babc1249c79aa840de94fa4a85bf55a453c813),multi_a(3,@0/270028'/0'/8'/
0'/<0;1>/*,@1/270028'/0'/8'/0'/<0;1>/*,@2/270028'/0'/8'/0'/<0;1>/*)),{and_v(v:older(32768),and_v(v:sha256(9ce09a8df1d1f8bc88
92441a9eb30d0a18bc7db81bb0fb3f54c6d9064e7b76ad),multi_a(2,@3/270028'/0'/8'/0'/<0;1>/*,@4/270028'/0'/8'/0'/<0;1>/*))),{and_v
(v:after(1173520),pk(@5/270028'/0'/8'/0'/<0;1>/*)),and_v(v:after(1383520),sha256(4743d7c47df21d29e3ed3dfec5d0c0a884ccc270863
7dddf771c36d214056954))}}}) --group-size 0 --experimental --fingerprint @0=39ec1b6e --fingerprint @1=d02bcedf --fingerprint
@2=5fd78eb3 --fingerprint @3=92c4c8b2 --fingerprint @4=086c047e --fingerprint @5=cef8224c
chunk-set-id: 0xec2a9
md1fas4fzqpp2040veppfzzqqvpc4xdw7mc9j32q9lm9qnfmufruzh27pyjw8n25ys5e6g4vx8jkqd
md1fas4fzqgx7jnay4pdl2kj98jqnyppq2z5e4cqqqsqqye4m88qn2xlr50chjyfyk7t0l0yw9prny
md1fas4fzqjyr20txrg2rz78mwqmkran74xxmyryu7mk45szzuz5e4kqq3aqgz4fjuhtu99yrl935q
md1fas4fzqadsq9guvp636r6lz8musa98376007chgvp2yyenp8pp3hmh0hw8pk6gnah9vug436nmt
md1fas4fzppgptf2sxxjqu7cxmw8gzhnklf0a0r4nwfvfj9jsyxcpr7h80sgjvqqkycjn9c66qq9z
warning: --experimental relaxed the signature rule. This descriptor has at least one spend path that needs NO key, so whoeve
r learns its preimage can spend it alone. If that preimage is engraved, THE PLATE IS BEARER ACCESS. Malleability, resource l
imits, repeated keys and timelock mixing were still checked.
note: stdout is a keyless descriptor template (no keys)
[exit 0]

The keyless template is 5 md1 chunk(s).

$ /scratch/code/shibboleth/descriptor-mnemonic/target/debug/md verify --template tr(50929b74c1a04954b78b4b6035e97a5e078a5a0f
28ec96d547bfee9ace803ac0,{and_v(v:sha256(ede0b28a805feca09a77c48f82babc1249c79aa840de94fa4a85bf55a453c813),multi_a(3,@0/2700
28'/0'/8'/0'/<0;1>/*,@1/270028'/0'/8'/0'/<0;1>/*,@2/270028'/0'/8'/0'/<0;1>/*)),{and_v(v:older(32768),and_v(v:sha256(9ce09a8d
f1d1f8bc8892441a9eb30d0a18bc7db81bb0fb3f54c6d9064e7b76ad),multi_a(2,@3/270028'/0'/8'/0'/<0;1>/*,@4/270028'/0'/8'/0'/<0;1>/
*))),{and_v(v:after(1173520),pk(@5/270028'/0'/8'/0'/<0;1>/*)),and_v(v:after(1383520),sha256(4743d7c47df21d29e3ed3dfec5d0c0a8
84ccc2708637dddf771c36d214056954))}}}) --fingerprint @0=39ec1b6e --fingerprint @1=d02bcedf --fingerprint @2=5fd78eb3 --finge
rprint @3=92c4c8b2 --fingerprint @4=086c047e --fingerprint @5=cef8224c md1fas4fzqpp2040veppfzzqqvpc4xdw7mc9j32q9lm9qnfmufruz
h27pyjw8n25ys5e6g4vx8jkqd md1fas4fzqgx7jnay4pdl2kj98jqnyppq2z5e4cqqqsqqye4m88qn2xlr50chjyfyk7t0l0yw9prny md1fas4fzqjyr20txrg
2rz78mwqmkran74xxmyryu7mk45szzuz5e4kqq3aqgz4fjuhtu99yrl935q md1fas4fzqadsq9guvp636r6lz8musa98376007chgvp2yyenp8pp3hmh0hw8pk6
gnah9vug436nmt md1fas4fzppgptf2sxxjqu7cxmw8gzhnklf0a0r4nwfvfj9jsyxcpr7h80sgjvqqkycjn9c66qq9z --experimental
OK
[exit 0]

$ /scratch/code/shibboleth/descriptor-mnemonic/target/debug/md inspect md1fas4fzqpp2040veppfzzqqvpc4xdw7mc9j32q9lm9qnfmufruz
h27pyjw8n25ys5e6g4vx8jkqd md1fas4fzqgx7jnay4pdl2kj98jqnyppq2z5e4cqqqsqqye4m88qn2xlr50chjyfyk7t0l0yw9prny md1fas4fzqjyr20txrg
2rz78mwqmkran74xxmyryu7mk45szzuz5e4kqq3aqgz4fjuhtu99yrl935q md1fas4fzqadsq9guvp636r6lz8musa98376007chgvp2yyenp8pp3hmh0hw8pk6
gnah9vug436nmt md1fas4fzppgptf2sxxjqu7cxmw8gzhnklf0a0r4nwfvfj9jsyxcpr7h80sgjvqqkycjn9c66qq9z
… clipped; full text in transcript_rcw.txt
```

```
$ /scratch/code/shibboleth/descriptor-mnemonic/target/debug/md encode tr(50929b74c1a04954b78b4b6035e97a5e078a5a0f28ec96d547b
fee9ace803ac0,{and_v(v:sha256(ede0b28a805feca09a77c48f82babc1249c79aa840de94fa4a85bf55a453c813),multi_a(3,@0/270028'/0'/8'/
0'/<0;1>/*,@1/270028'/0'/8'/0'/<0;1>/*,@2/270028'/0'/8'/0'/<0;1>/*)),{and_v(v:older(32768),and_v(v:sha256(9ce09a8df1d1f8bc88
92441a9eb30d0a18bc7db81bb0fb3f54c6d9064e7b76ad),multi_a(2,@3/270028'/0'/8'/0'/<0;1>/*,@4/270028'/0'/8'/0'/<0;1>/*))),{and_v
(v:after(1173520),pk(@5/270028'/0'/8'/0'/<0;1>/*)),and_v(v:after(1383520),sha256(4743d7c47df21d29e3ed3dfec5d0c0a884ccc270863
7dddf771c36d214056954))}}}) --key @0=xpub6Dn2FdZCkR6b6nCFWKMZraCwEcC7aedkRrVHf8DK5gSBrny6YJkiUZp9bL8j2a4osHSuhDsk1FBnSS81C82
qvZ4uiDofJGQKmNmmxqfUxCr --key @1=xpub6DdqV8W2dzVkyfWmNUJHkoWdH6JctbyGvKrJVUDUL1hSoUBGtfB3JtXoyYtvkVR3S6jyzagALehL56vqwcjXAc
5hoaCoL6yyrtCLQf1dg2f --key @2=xpub6FMyX6RDPGTJgt3hYThGymmhRUhntuA9HagmHmDPUm11hDKk6HLNFspSyRWqqdiSQYMQujjsL2Xn9ERJE9YDZR9jG
2X4UJbx53XE3b8Z5Bw --key @3=xpub6E9jU3J1VtuNb5GVBatKMhYkHiMM7dD6yK2x4fhDDSghXHQLxJQgFZnwLEXWc1jAb5ZCsNUhQGDqUdy3Jt3kRmSMG6Fn
yBnM1PRXmD6bXb4 --key @4=xpub6Ep7XhWjqPzjf5HTCQTf8M8QRYUqGHJLpXQ9qbU7xJrGGCsG9Fba4QUtughDTBLJgf5praqaxKYUhmTvKMsWiG8Gs24jJgL
zspG2tstkBdz --key @5=xpub6EvqkLWC3QnMavFkcHB5Kzfnv8FeikdNwurio9zyTWgNnwdsZXf7KdcGAh9to36zwLckiVkLpyjBUxD9f9xWSuT9KUmw1v1Qqz
qDuYuBW7q --fingerprint @0=39ec1b6e --fingerprint @1=d02bcedf --fingerprint @2=5fd78eb3 --fingerprint @3=92c4c8b2 --fingerpr
int @4=086c047e --fingerprint @5=cef8224c --group-size 0 --experimental
chunk-set-id: 0x30662
md1fxpnz8qpp2040veppfzzqqvpc4xdw7mc9j32q9lm9qnfmufruzh27pyjw8n25yph55lfqkpz9ct5hzvhaa
md1fxpnz8q259ha26g57gzvsyypg2nxhqqqzqqqnxhvuuzdgmuw3lz7g3yjyr20txrg2rz7q20ds83ekhpw3k
md1fxpnz8qnahqdmp7el2nrdjpjw0dm26gppwp2v6mqqg7syp25e4kqq4r3s828g0tugl0jqhfr09j9kxvddt
md1fxpnz8qe620ra57la3wscz5gfnxzwzrr0hwlwuwrd5s5q454grrfqw0vrdhr5ptem05hs6kp79jrp43ync
md1fxpnz8p8tcavmjtzv3v5ppkqgl4emuzynqkvggp0kjhjdndqdhxe88vw6pfzlczdf76zqq8mkufmrwtcfn
md1fxpnz8pve9q6stndtvgpmda6h50kcp65nezrrjmp94j9lfy6ns7qncgukzhpldcfwszgqw56mlrxmkpsjk
md1fxpnz8p3xw7332s5z7wjcjlvuel7ksrw329kvdzpqmetn093g7teztd7e9c664686ljgsynqk65spm59fc
md1fxpnz8pm0wqgpnnzt6etcxuqw4cjs9jldx35ews7ldpzqu4eegntc642f0fysud9jryjsxlwv9tvuptmt6
md1fxpnz8zq4cvq5c66qznwuz8jvayr34rapk22tpkmrwxxyplpm0v28prczuppm3ewkk3js08d5zydcajz46
md1fxpnz8zdqr46wjjqudcfuqn5d4mektc0jduujqxg70f5hn7rpz50u5sy7wjaftmvss39qt2k7n8ccqemc9
md1fxpnz8zh9cenm4ksx7lte9zpnszutezsrdjq0kgzgma9nugkut9zm882y22v6wuv0vdwsvr5hsx8m2qf2h
md1fxpnz8zugfngxa4uzashfzf9u63ryvvyrcwpy9echwgsdc4cvutz3692mfhe5d3l95lksecrfstkjzhkzd
md1fxpnz8rqz6umdpjgyfzz5qr9d2auqe8gpeghq97ejy8jvvqp7ekdlm9203gz5py08czsqaurwjghlh5hs2
md1fxpnz8r0960hu9xmjh3rzxefuaz5u44smg3esd4vtty488c6k40mzcmdwz3g4gun7qx3s5xlrarft6uwdc
md1fxpnz8r597zah0lp6xaxlykwc49z08l3jk3hkj9tjtgc3kwjdjcgu6t5aqqxpqu470znf8hs
warning: --experimental relaxed the signature rule. This descriptor has at least one spend path that needs NO key, so whoeve
r learns its preimage can spend it alone. If that preimage is engraved, THE PLATE IS BEARER ACCESS. Malleability, resource l
imits, repeated keys and timelock mixing were still checked.
note: stdout is watch-only — public keys only, cannot spend
```

```
[exit 0]

The keyed card set is 15 md1 chunk(s).

  wallet-descriptor-template-id (keyless): 68a1a888385797337ce5debc90fcfb1e
  wallet-descriptor-template-id (keyed)  : 68a1a888385797337ce5debc90fcfb1e
  wallet-policy-id              (keyed)  : 5c2c6fb54ce0c9042f9619dbe659472e

ONE WALLET, TWO IDS. The template id is key-STABLE and identical either
way; the policy id depends on the keys. A card binds to whichever of the
two matches its own form.
```

tr — descriptor and addresses

```
$ /scratch/code/shibboleth/descriptor-mnemonic/target/debug/md descriptor md1fxpnz8qpp2040veppfzzqqvpc4xdw7mc9j32q9lm9qnfmuf
ruzh27pyjw8n25yph55lfqkpz9ct5hzvhaa md1fxpnz8q259ha26g57gzvsyypg2nxhqqqzqqqnxhvuuzdgmuw3lz7g3yjyr20txrg2rz7q20ds83ekhpw3k md
1fxpnz8qnahqdmp7el2nrdjpjw0dm26gppwp2v6mqqg7syp25e4kqq4r3s828g0tugl0jqhfr09j9kxvddt md1fxpnz8qe620ra57la3wscz5gfnxzwzrr0hwlw
uwrd5s5q454grrfqw0vrdhr5ptem05hs6kp79jrp43ync md1fxpnz8p8tcavmjtzv3v5ppkqgl4emuzynqkvggp0kjhjdndqdhxe88vw6pfzlczdf76zqq8mkuf
mrwtcfn md1fxpnz8pve9q6stndtvgpmda6h50kcp65nezrrjmp94j9lfy6ns7qncgukzhpldcfwszgqw56mlrxmkpsjk md1fxpnz8p3xw7332s5z7wjcjlvuel
7ksrw329kvdzpqmetn093g7teztd7e9c664686ljgsynqk65spm59fc md1fxpnz8pm0wqgpnnzt6etcxuqw4cjs9jldx35ews7ldpzqu4eegntc642f0fysud9j
ryjsxlwv9tvuptmt6 md1fxpnz8zq4cvq5c66qznwuz8jvayr34rapk22tpkmrwxxyplpm0v28prczuppm3ewkk3js08d5zydcajz46 md1fxpnz8zdqr46wjjqu
dcfuqn5d4mektc0jduujqxg70f5hn7rpz50u5sy7wjaftmvss39qt2k7n8ccqemc9 md1fxpnz8zh9cenm4ksx7lte9zpnszutezsrdjq0kgzgma9nugkut9zm88
2y22v6wuv0vdwsvr5hsx8m2qf2h md1fxpnz8zugfngxa4uzashfzf9u63ryvvyrcwpy9echwgsdc4cvutz3692mfhe5d3l95lksecrfstkjzhkzd md1fxpnz8r
qz6umdpjgyfzz5qr9d2auqe8gpeghq97ejy8jvvqp7ekdlm9203gz5py08czsqaurwjghlh5hs2 md1fxpnz8r0960hu9xmjh3rzxefuaz5u44smg3esd4vtty48
8c6k40mzcmdwz3g4gun7qx3s5xlrarft6uwdc md1fxpnz8r597zah0lp6xaxlykwc49z08l3jk3hkj9tjtgc3kwjdjcgu6t5aqqxpqu470znf8hs
tr(50929b74c1a04954b78b4b6035e97a5e078a5a0f28ec96d547bfee9ace803ac0,{and_v(v:sha256(ede0b28a805feca09a77c48f82babc1249c79aa8
40de94fa4a85bf55a453c813),multi_a(3,[39ec1b6e/270028'/0'/8'/0']xpub661MyMwAqRbcEnD2AtBLZFdpjyX4eUjLYHGbAJM5wUaNHiMgaovDDvj6k
jxx987aSzyKbmAo6nsse35VZ4hEW3ya1R1Kj5oSiMEzmYHYCYH/<0;1>/*,[d02bcedf/270028'/0'/8'/0']xpub661MyMwAqRbcF1ymCBSeBwHofzVWnLXr3p
L4oc8CBvecuCrmMCKSQwxcpSUQUpQcP9A7JRwms5aQsW8DEXGJjXBDX76ke8WXQ6aVtAJP7HX/<0;1>/*,[5fd78eb3/270028'/0'/8'/0']xpub661MyMwAqRb
cEnxHs9XNhi1LetXt61kvKxw1eYtjtrHYrysh3wyxoxUaNxG8TDehBthM3J5R7CHnZS6gdT8PY3kY3PJqQCyriq4swVEAg57/<0;1>/*)),{and_v(v:older(32
768),and_v(v:sha256(9ce09a8df1d1f8bc8892441a9eb30d0a18bc7db81bb0fb3f54c6d9064e7b76ad),multi_a(2,[92c4c8b2/270028'/0'/8'/0']x
pub661MyMwAqRbcEdSBbQ28dKzUUygJkaMomhRdmcB4fXrrsZBWAwJxCFDFsVLfQn8HdBXqNMcAgHKozqgjqy6aDKcVMYsjrhNYLxp49VeJj8y/<0;1>/*,[086c
047e/270028'/0'/8'/0']xpub661MyMwAqRbcEsTQ3DwNCxRfdaVqoMP4FN1HqB4vHqRbxABdmsTKHLX2J4nJiDwExnr32u7xS5KstfjUq3LmzteztWRCMpdtT3
FBNXX7zvS/<0;1>/*))),{and_v(v:after(1173520),pk([cef8224c/270028'/0'/8'/0']xpub661MyMwAqRbcF68jPjC8VwRUTZfaqP4YEMwX2f3RT7TBH
1hXzC6rYwDNyekvEj49rowooqTSvqQzmC3z7gkD6547TYCHZ1MNcwSsmktWDx7/<0;1>/*)),and_v(v:after(1383520),sha256(4743d7c47df21d29e3ed3
dfec5d0c0a884ccc2708637dddf771c36d214056954))}}})#h64g8jju
note: stdout is watch-only — public keys only, cannot spend
[exit 0]

  1299 bytes
```

Five receive and five change addresses, on the host (5 and 5 derived):

```
$ /scratch/code/shibboleth/descriptor-mnemonic/target/debug/md address md1fxpnz8qpp2040veppfzzqqvpc4xdw7mc9j32q9lm9qnfmufruz
h27pyjw8n25yph55lfqkpz9ct5hzvhaa md1fxpnz8q259ha26g57gzvsyypg2nxhqqqzqqqnxhvuuzdgmuw3lz7g3yjyr20txrg2rz7q20ds83ekhpw3k md1fx
pnz8qnahqdmp7el2nrdjpjw0dm26gppwp2v6mqqg7syp25e4kqq4r3s828g0tugl0jqhfr09j9kxvddt md1fxpnz8qe620ra57la3wscz5gfnxzwzrr0hwlwuwr
d5s5q454grrfqw0vrdhr5ptem05hs6kp79jrp43ync md1fxpnz8p8tcavmjtzv3v5ppkqgl4emuzynqkvggp0kjhjdndqdhxe88vw6pfzlczdf76zqq8mkufmrw
tcfn md1fxpnz8pve9q6stndtvgpmda6h50kcp65nezrrjmp94j9lfy6ns7qncgukzhpldcfwszgqw56mlrxmkpsjk md1fxpnz8p3xw7332s5z7wjcjlvuel7ks
rw329kvdzpqmetn093g7teztd7e9c664686ljgsynqk65spm59fc md1fxpnz8pm0wqgpnnzt6etcxuqw4cjs9jldx35ews7ldpzqu4eegntc642f0fysud9jryj
sxlwv9tvuptmt6 md1fxpnz8zq4cvq5c66qznwuz8jvayr34rapk22tpkmrwxxyplpm0v28prczuppm3ewkk3js08d5zydcajz46 md1fxpnz8zdqr46wjjqudcf
uqn5d4mektc0jduujqxg70f5hn7rpz50u5sy7wjaftmvss39qt2k7n8ccqemc9 md1fxpnz8zh9cenm4ksx7lte9zpnszutezsrdjq0kgzgma9nugkut9zm882y2
2v6wuv0vdwsvr5hsx8m2qf2h md1fxpnz8zugfngxa4uzashfzf9u63ryvvyrcwpy9echwgsdc4cvutz3692mfhe5d3l95lksecrfstkjzhkzd md1fxpnz8rqz6
umdpjgyfzz5qr9d2auqe8gpeghq97ejy8jvvqp7ekdlm9203gz5py08czsqaurwjghlh5hs2 md1fxpnz8r0960hu9xmjh3rzxefuaz5u44smg3esd4vtty488c6
k40mzcmdwz3g4gun7qx3s5xlrarft6uwdc md1fxpnz8r597zah0lp6xaxlykwc49z08l3jk3hkj9tjtgc3kwjdjcgu6t5aqqxpqu470znf8hs --chain 0 --c
ount 5
bc1p8h77a4mgl49mld6crpdpvx4jsg6gclkv9vr93enl54cdz0gp6v8smuqr2n
bc1panm4lv4hl8n96xccufjc7rkd4hhuvled2hp5eqdhncaqvpuvvfdq5z9t8t
bc1pf0j39ep28fqssev4xf6mrtnjp5c896qgklwu5tw2lvqcvg7jzxxs63pwxu
bc1psz5533c3n6agpt48lr3x9ugk77zlftcfug03upeh74fzjtchfypqnnljqn
bc1prf5sadp5390xv2n0famzsc836k8alxqqr9r7sfzxyn6q6zrpp63sw9ym0w
note: stdout is watch-only — public keys only, cannot spend
[exit 0]

$ /scratch/code/shibboleth/descriptor-mnemonic/target/debug/md address md1fxpnz8qpp2040veppfzzqqvpc4xdw7mc9j32q9lm9qnfmufruz
h27pyjw8n25yph55lfqkpz9ct5hzvhaa md1fxpnz8q259ha26g57gzvsyypg2nxhqqqzqqqnxhvuuzdgmuw3lz7g3yjyr20txrg2rz7q20ds83ekhpw3k md1fx
pnz8qnahqdmp7el2nrdjpjw0dm26gppwp2v6mqqg7syp25e4kqq4r3s828g0tugl0jqhfr09j9kxvddt md1fxpnz8qe620ra57la3wscz5gfnxzwzrr0hwlwuwr
d5s5q454grrfqw0vrdhr5ptem05hs6kp79jrp43ync md1fxpnz8p8tcavmjtzv3v5ppkqgl4emuzynqkvggp0kjhjdndqdhxe88vw6pfzlczdf76zqq8mkufmrw
tcfn md1fxpnz8pve9q6stndtvgpmda6h50kcp65nezrrjmp94j9lfy6ns7qncgukzhpldcfwszgqw56mlrxmkpsjk md1fxpnz8p3xw7332s5z7wjcjlvuel7ks
rw329kvdzpqmetn093g7teztd7e9c664686ljgsynqk65spm59fc md1fxpnz8pm0wqgpnnzt6etcxuqw4cjs9jldx35ews7ldpzqu4eegntc642f0fysud9jryj
sxlwv9tvuptmt6 md1fxpnz8zq4cvq5c66qznwuz8jvayr34rapk22tpkmrwxxyplpm0v28prczuppm3ewkk3js08d5zydcajz46 md1fxpnz8zdqr46wjjqudcf
uqn5d4mektc0jduujqxg70f5hn7rpz50u5sy7wjaftmvss39qt2k7n8ccqemc9 md1fxpnz8zh9cenm4ksx7lte9zpnszutezsrdjq0kgzgma9nugkut9zm882y2
2v6wuv0vdwsvr5hsx8m2qf2h md1fxpnz8zugfngxa4uzashfzf9u63ryvvyrcwpy9echwgsdc4cvutz3692mfhe5d3l95lksecrfstkjzhkzd md1fxpnz8rqz6
umdpjgyfzz5qr9d2auqe8gpeghq97ejy8jvvqp7ekdlm9203gz5py08czsqaurwjghlh5hs2 md1fxpnz8r0960hu9xmjh3rzxefuaz5u44smg3esd4vtty488c6
k40mzcmdwz3g4gun7qx3s5xlrarft6uwdc md1fxpnz8r597zah0lp6xaxlykwc49z08l3jk3hkj9tjtgc3kwjdjcgu6t5aqqxpqu470znf8hs --chain 1 --c
ount 5
bc1pt7g96plv4mw73vrs64wvq2eswq9zrh4r0ztnx7huuma94xzskkhqm8vkn4
bc1pp3fpcxm9zkp75hguhj5qh804wxzrff5tp6wcunr0up7j3xp46v3su0mntq
bc1pg6urxdm50ujxw8kl7zjl7d2nwh75hxl4hhpjalzzxx7pv2ewuy9qcctund
bc1pf8rf0ylce0g0dd4k44w055g4jnywsh7mnd7ysnqt4xmy9fqtw6ws0l38ps
bc1p80t9jzls9624gttgq8tw65q7jlvzk395679qh77pj3quv02uu3dq6xm86k
note: stdout is watch-only — public keys only, cannot spend
[exit 0]
```

wsh — the cards

```
Fingerprints are declared (F-227): all six slots share one origin path,
so without them a gathered key card matches every slot and seating must
refuse the whole set.

$ /scratch/code/shibboleth/descriptor-mnemonic/target/debug/md encode wsh(or_i(and_v(v:sha256(ede0b28a805feca09a77c48f82babc
1249c79aa840de94fa4a85bf55a453c813),multi(3,@0/270028'/0'/9'/1'/<0;1>/*,@1/270028'/0'/9'/1'/<0;1>/*,@2/270028'/0'/9'/1'/<0;1
>/*)),or_i(and_v(v:older(32768),and_v(v:sha256(9ce09a8df1d1f8bc8892441a9eb30d0a18bc7db81bb0fb3f54c6d9064e7b76ad),multi(2,@3/
270028'/0'/9'/1'/<0;1>/*,@4/270028'/0'/9'/1'/<0;1>/*))),or_i(and_v(v:after(1173520),pk(@5/270028'/0'/9'/1'/<0;1>/*)),and_v
(v:after(1383520),sha256(4743d7c47df21d29e3ed3dfec5d0c0a884ccc2708637dddf771c36d214056954)))))) --group-size 0 --experimenta
l --fingerprint @0=39ec1b6e --fingerprint @1=d02bcedf --fingerprint @2=5fd78eb3 --fingerprint @3=92c4c8b2 --fingerprint @4=0
86c047e --fingerprint @5=cef8224c
chunk-set-id: 0xf2bf0
md1f72lszqpp2040veppfx8qqxpx2v6aahst9z5qtlk2pxnhcj8c9w4uzfyu0x4ggmed54w92q28q4
md1f72lszqgda986f2zm74dy20ypxxzzq5efntsqqpqqqfntkwwpx5d78gl30ygjgsle764dhvdvgr
md1f72lszq5gx57kvxs5x9u0kuphv8m8a2vdkgxfeahdtgcy9cefntvqpr6qs92nxl9npextjdsvkn
md1f72lszq6mqq23ccr4r5847y0hep620ra57la3wscz5gfnxzwzrr0hwlwuwrd5s5pa2we2d8dteg
md1f72lszpzszkj4qvdypeasdkuws908d7jl678txujcnyt9qgdsz8awwlq3ycqqn0sdc05cheg5a
warning: --experimental relaxed the signature rule. This descriptor has at least one spend path that needs NO key, so whoeve
r learns its preimage can spend it alone. If that preimage is engraved, THE PLATE IS BEARER ACCESS. Malleability, resource l
imits, repeated keys and timelock mixing were still checked.
note: stdout is a keyless descriptor template (no keys)
[exit 0]

The keyless template is 5 md1 chunk(s).

$ /scratch/code/shibboleth/descriptor-mnemonic/target/debug/md verify --template wsh(or_i(and_v(v:sha256(ede0b28a805feca09a7
7c48f82babc1249c79aa840de94fa4a85bf55a453c813),multi(3,@0/270028'/0'/9'/1'/<0;1>/*,@1/270028'/0'/9'/1'/<0;1>/*,@2/270028'/
0'/9'/1'/<0;1>/*)),or_i(and_v(v:older(32768),and_v(v:sha256(9ce09a8df1d1f8bc8892441a9eb30d0a18bc7db81bb0fb3f54c6d9064e7b76a
d),multi(2,@3/270028'/0'/9'/1'/<0;1>/*,@4/270028'/0'/9'/1'/<0;1>/*))),or_i(and_v(v:after(1173520),pk(@5/270028'/0'/9'/1'/<0;
1>/*)),and_v(v:after(1383520),sha256(4743d7c47df21d29e3ed3dfec5d0c0a884ccc2708637dddf771c36d214056954)))))) --fingerprint @0
=39ec1b6e --fingerprint @1=d02bcedf --fingerprint @2=5fd78eb3 --fingerprint @3=92c4c8b2 --fingerprint @4=086c047e --fingerpr
int @5=cef8224c md1f72lszqpp2040veppfx8qqxpx2v6aahst9z5qtlk2pxnhcj8c9w4uzfyu0x4ggmed54w92q28q4 md1f72lszqgda986f2zm74dy20ypx
xzzq5efntsqqpqqqfntkwwpx5d78gl30ygjgsle764dhvdvgr md1f72lszq5gx57kvxs5x9u0kuphv8m8a2vdkgxfeahdtgcy9cefntvqpr6qs92nxl9npextjd
svkn md1f72lszq6mqq23ccr4r5847y0hep620ra57la3wscz5gfnxzwzrr0hwlwuwrd5s5pa2we2d8dteg md1f72lszpzszkj4qvdypeasdkuws908d7jl678t
xujcnyt9qgdsz8awwlq3ycqqn0sdc05cheg5a --experimental
OK
[exit 0]

$ /scratch/code/shibboleth/descriptor-mnemonic/target/debug/md inspect md1f72lszqpp2040veppfx8qqxpx2v6aahst9z5qtlk2pxnhcj8c9
w4uzfyu0x4ggmed54w92q28q4 md1f72lszqgda986f2zm74dy20ypxxzzq5efntsqqpqqqfntkwwpx5d78gl30ygjgsle764dhvdvgr md1f72lszq5gx57kvxs
5x9u0kuphv8m8a2vdkgxfeahdtgcy9cefntvqpr6qs92nxl9npextjdsvkn md1f72lszq6mqq23ccr4r5847y0hep620ra57la3wscz5gfnxzwzrr0hwlwuwrd5
s5pa2we2d8dteg md1f72lszpzszkj4qvdypeasdkuws908d7jl678txujcnyt9qgdsz8awwlq3ycqqn0sdc05cheg5a
… clipped; full text in transcript_rcw.txt
```

```
$ /scratch/code/shibboleth/descriptor-mnemonic/target/debug/md encode wsh(or_i(and_v(v:sha256(ede0b28a805feca09a77c48f82babc
1249c79aa840de94fa4a85bf55a453c813),multi(3,@0/270028'/0'/9'/1'/<0;1>/*,@1/270028'/0'/9'/1'/<0;1>/*,@2/270028'/0'/9'/1'/<0;1
>/*)),or_i(and_v(v:older(32768),and_v(v:sha256(9ce09a8df1d1f8bc8892441a9eb30d0a18bc7db81bb0fb3f54c6d9064e7b76ad),multi(2,@3/
270028'/0'/9'/1'/<0;1>/*,@4/270028'/0'/9'/1'/<0;1>/*))),or_i(and_v(v:after(1173520),pk(@5/270028'/0'/9'/1'/<0;1>/*)),and_v
(v:after(1383520),sha256(4743d7c47df21d29e3ed3dfec5d0c0a884ccc2708637dddf771c36d214056954)))))) --key @0=xpub6EddmgK6uMbst72
51zES359MJzfz3o2wTJWeefTzSiPJf1FUg6easA2Uk3jUbztBffWS4Dg2oWjhomvU2jJKr4EZukDfxVxb4VJva3jXGAK --key @1=xpub6EPimu5ztRjy6dfECv
b9AQNAadNHN8qumZYQhvdPWh9U9xJ9XMi328aXAFMGHwc11wZWrmdwsHUWJn5in6BHywGrbJx5ZLUWe8xhNPRf5kL --key @2=xpub6EVA2mZiCBtGYbGM5eEs4
2k8tso4QekUog2t8U14UDyjUYpKByLYcXB8t3yVAvapQymRSK2MDM84cspZa5udQEEUSVGo28RJJt47eeC846C --key @3=xpub6DZroUpPBJ8ZSJCj4DP2t3Cf
D8vESCKb1Y4C6LgU364QkX3tRVUfuUERm8FGifHGXkSNEALDMXgDJzVKDg79usbEoBF56Tf3gyNFQ4TjrHq --key @4=xpub6FFo9cyULG6UEz8usJ7R8JQjvwE
saFusL4jv2WUxsfJm4TPXR6zUpsdKg4MofE179in7i1qKFJK1CsweWSntz2FW2tLYSiY1uYpjh9ataZa --key @5=xpub6DjFP13A3XC9SPE6XnxQ2SKsF1Sr1K
bjcNCEKfABA3WAmGiECo8HNqfU71dmPoQshBGqzubYhvgiSF44qHUh64WnM8Z42qdf2fDKMCPS7eX --fingerprint @0=39ec1b6e --fingerprint @1=d02
bcedf --fingerprint @2=5fd78eb3 --fingerprint @3=92c4c8b2 --fingerprint @4=086c047e --fingerprint @5=cef8224c --group-size 0
--experimental
chunk-set-id: 0xa4ff4
md1f5nl58qpp2040veppfx8qqxpx2v6aahst9z5qtlk2pxnhcj8c9w4uzfyu0x4ggr0ff7jqeh2qcmjtmnm68
md1f5nl58qdgt06453fusyccggzn9xdwqqqyqqpxdweecy63hcar79u3zfygx57kvxs5x9uqf93jqxfl9ppns
md1f5nl58qhmwqmkran74xxmyryu7mk45vzzuv5e4kqq3aqgz4fntvqp28rqw5ws7hc37lyqg66puucsn7wgk
md1f5nl58qm5578mfalmzaps9gsnxvyuyxxlwa7acuxmfpgptf2sxxjqu7cxmw8gzhnklf0sz7u90d6z6x4vj
md1f5nl58pxh36ehykyezegzrvq3ltnhcyfxpvcss78u59v8gsu55wj3crtt4vs33fy3jupswd3pht0yqkvmr
md1f5nl58pd6uq0h2z2mwrxjfvd9tg0crh8n4t07049utsul4w5dyeanq0c5j5e0v2yeazrswfymhg943ez9q
md1f5nl58ph3xdp7jdwd4x23agzgmtyeta0sgkvqq005enx2qvg99s9s4zypw88eyeyvszvszk2mu8es7jcj3
md1f5nl58plxfygzzdkmhugqfedcfejakelta7k97h5hapszwrfazruxtax3rqls29j4mj2q94ztc62lf0f9p
md1f5nl58zpjrhst0rkv9m9e77su2nf9sgxcdxq2lqewdp0jr30af03vhes8kuf8wsfhhs7qxv0acr9njq92n
md1f5nl58zd24tpdkh3prfu7f47g4ntt488aal33v7xgtm7nzyxeyt99rm5cmeavkayw4e0s24a9vcq6xq8de
md1f5nl58z4nmhnfk2h56yhptk4lf27gvugwccqqaqx5vfn7xcy7t7l04mrquhq5k7vj0ywqmrddyhf20urlg
md1f5nl58z6t25tltp3ckcyejn3mjr868cj73pd0pw3euqyg9glwyfhcsvvsp9u8hdqmfh3syex2yyjd4kn6f
md1f5nl58rxpc8xhhu3sqxhwsyzw9jr32lcjrputwmdtlut5elkhtynnq7tm8drts47t2qkqmx3ulr0fwgpkm
md1f5nl58r2vnzc02aqrgcupzytm7nek6p8gwvfmg27uwjrp92q0n77qtrsyu48ualq5q0vsjq0c8zssp3w30
md1f5nl58r3ujfumqccaemfyfg3cqu633yeh80klhqknfyd4kvjsnwn05ng0clf9h7z2z4ecw
warning: --experimental relaxed the signature rule. This descriptor has at least one spend path that needs NO key, so whoeve
r learns its preimage can spend it alone. If that preimage is engraved, THE PLATE IS BEARER ACCESS. Malleability, resource l
imits, repeated keys and timelock mixing were still checked.
note: stdout is watch-only — public keys only, cannot spend
```

```
[exit 0]

The keyed card set is 15 md1 chunk(s).

  wallet-descriptor-template-id (keyless): daee67be4eacf85e8b832ae64fc06566
  wallet-descriptor-template-id (keyed)  : daee67be4eacf85e8b832ae64fc06566
  wallet-policy-id              (keyed)  : 174ee71427fcb8edcec5d14ae5989449

ONE WALLET, TWO IDS. The template id is key-STABLE and identical either
way; the policy id depends on the keys. A card binds to whichever of the
two matches its own form.
```

wsh — descriptor and addresses

```
$ /scratch/code/shibboleth/descriptor-mnemonic/target/debug/md descriptor md1f5nl58qpp2040veppfx8qqxpx2v6aahst9z5qtlk2pxnhcj
8c9w4uzfyu0x4ggr0ff7jqeh2qcmjtmnm68 md1f5nl58qdgt06453fusyccggzn9xdwqqqyqqpxdweecy63hcar79u3zfygx57kvxs5x9uqf93jqxfl9ppns md
1f5nl58qhmwqmkran74xxmyryu7mk45vzzuv5e4kqq3aqgz4fntvqp28rqw5ws7hc37lyqg66puucsn7wgk md1f5nl58qm5578mfalmzaps9gsnxvyuyxxlwa7a
cuxmfpgptf2sxxjqu7cxmw8gzhnklf0sz7u90d6z6x4vj md1f5nl58pxh36ehykyezegzrvq3ltnhcyfxpvcss78u59v8gsu55wj3crtt4vs33fy3jupswd3pht
0yqkvmr md1f5nl58pd6uq0h2z2mwrxjfvd9tg0crh8n4t07049utsul4w5dyeanq0c5j5e0v2yeazrswfymhg943ez9q md1f5nl58ph3xdp7jdwd4x23agzgmt
yeta0sgkvqq005enx2qvg99s9s4zypw88eyeyvszvszk2mu8es7jcj3 md1f5nl58plxfygzzdkmhugqfedcfejakelta7k97h5hapszwrfazruxtax3rqls29j4
mj2q94ztc62lf0f9p md1f5nl58zpjrhst0rkv9m9e77su2nf9sgxcdxq2lqewdp0jr30af03vhes8kuf8wsfhhs7qxv0acr9njq92n md1f5nl58zd24tpdkh3p
rfu7f47g4ntt488aal33v7xgtm7nzyxeyt99rm5cmeavkayw4e0s24a9vcq6xq8de md1f5nl58z4nmhnfk2h56yhptk4lf27gvugwccqqaqx5vfn7xcy7t7l04m
rquhq5k7vj0ywqmrddyhf20urlg md1f5nl58z6t25tltp3ckcyejn3mjr868cj73pd0pw3euqyg9glwyfhcsvvsp9u8hdqmfh3syex2yyjd4kn6f md1f5nl58r
xpc8xhhu3sqxhwsyzw9jr32lcjrputwmdtlut5elkhtynnq7tm8drts47t2qkqmx3ulr0fwgpkm md1f5nl58r2vnzc02aqrgcupzytm7nek6p8gwvfmg27uwjrp
92q0n77qtrsyu48ualq5q0vsjq0c8zssp3w30 md1f5nl58r3ujfumqccaemfyfg3cqu633yeh80klhqknfyd4kvjsnwn05ng0clf9h7z2z4ecw
wsh(or_i(and_v(v:sha256(ede0b28a805feca09a77c48f82babc1249c79aa840de94fa4a85bf55a453c813),multi(3,[39ec1b6e/270028'/0'/9'/
1']xpub661MyMwAqRbcGxALkZqYHiTZg3UgzcdVxNzbCBdBjT2ViaCZYexPy2fzQnVfu7Lw5DFYpmrTsp9nXydScuJhD3VEEJhhdoEDw2rEmq3qQ8s/<0;1>/*,
[d02bcedf/270028'/0'/9'/1']xpub661MyMwAqRbcGsZt7ZRaPrrBpNVojL35MWzsbBrXyMwtbrBLbr3yE2qeffvCXq9tkygXAidxdgtYYkiVVDHtyB9tdJMJ5
MFM58rTfnP92Je/<0;1>/*,[5fd78eb3/270028'/0'/9'/1']xpub661MyMwAqRbcGv2wz7ULSPNpQJbCYTLzd5mZPZVZpwhrYvTcQK6tgP5LLPtkDcfYYnDf7o
f1fc6iHeTZ6Zv9dw9kRMuq8WYZ7GVCeZBoX9f/<0;1>/*)),or_i(and_v(v:older(32768),and_v(v:sha256(9ce09a8df1d1f8bc8892441a9eb30d0a18b
c7db81bb0fb3f54c6d9064e7b76ad),multi(2,[92c4c8b2/270028'/0'/9'/1']xpub661MyMwAqRbcFYVZ1cCYQR8QLuWzvXycGjgW6VRM8oauwKFP94sa3h
ftUjqjjZtg7gqLcyRizTdk1mYtb7A2PSgUYyA9CBX8pg1pb8x5amq/<0;1>/*,[086c047e/270028'/0'/9'/1']xpub661MyMwAqRbcFJDPm1NKXvsQyiXHWHy
JXsx6rXViuwjr6Fb5PqpCHdwNst2YGBP8rWYCapMzqA82mCBdw8BWYtBuFdaw1JawqTrUnNtt1Mq/<0;1>/*))),or_i(and_v(v:after(1173520),pk([cef8
224c/270028'/0'/9'/1']xpub661MyMwAqRbcFhReYzdHhyfkEAgnn7MxQptt1gEWAgQz5gQZXwVt7hTA9nSicshD5hCGc5b4wj38gSuR7jAt4gEwguwjZnc7ji
cbbwzMd5a/<0;1>/*)),and_v(v:after(1383520),sha256(4743d7c47df21d29e3ed3dfec5d0c0a884ccc2708637dddf771c36d214056954))))))#aqe
ftg67
note: stdout is watch-only — public keys only, cannot spend
[exit 0]

  1243 bytes
```

Five receive and five change addresses, on the host (5 and 5 derived):

```
$ /scratch/code/shibboleth/descriptor-mnemonic/target/debug/md address md1f5nl58qpp2040veppfx8qqxpx2v6aahst9z5qtlk2pxnhcj8c9
w4uzfyu0x4ggr0ff7jqeh2qcmjtmnm68 md1f5nl58qdgt06453fusyccggzn9xdwqqqyqqpxdweecy63hcar79u3zfygx57kvxs5x9uqf93jqxfl9ppns md1f5
nl58qhmwqmkran74xxmyryu7mk45vzzuv5e4kqq3aqgz4fntvqp28rqw5ws7hc37lyqg66puucsn7wgk md1f5nl58qm5578mfalmzaps9gsnxvyuyxxlwa7acux
mfpgptf2sxxjqu7cxmw8gzhnklf0sz7u90d6z6x4vj md1f5nl58pxh36ehykyezegzrvq3ltnhcyfxpvcss78u59v8gsu55wj3crtt4vs33fy3jupswd3pht0yq
kvmr md1f5nl58pd6uq0h2z2mwrxjfvd9tg0crh8n4t07049utsul4w5dyeanq0c5j5e0v2yeazrswfymhg943ez9q md1f5nl58ph3xdp7jdwd4x23agzgmtyet
a0sgkvqq005enx2qvg99s9s4zypw88eyeyvszvszk2mu8es7jcj3 md1f5nl58plxfygzzdkmhugqfedcfejakelta7k97h5hapszwrfazruxtax3rqls29j4mj2
q94ztc62lf0f9p md1f5nl58zpjrhst0rkv9m9e77su2nf9sgxcdxq2lqewdp0jr30af03vhes8kuf8wsfhhs7qxv0acr9njq92n md1f5nl58zd24tpdkh3prfu
7f47g4ntt488aal33v7xgtm7nzyxeyt99rm5cmeavkayw4e0s24a9vcq6xq8de md1f5nl58z4nmhnfk2h56yhptk4lf27gvugwccqqaqx5vfn7xcy7t7l04mrqu
hq5k7vj0ywqmrddyhf20urlg md1f5nl58z6t25tltp3ckcyejn3mjr868cj73pd0pw3euqyg9glwyfhcsvvsp9u8hdqmfh3syex2yyjd4kn6f md1f5nl58rxpc
8xhhu3sqxhwsyzw9jr32lcjrputwmdtlut5elkhtynnq7tm8drts47t2qkqmx3ulr0fwgpkm md1f5nl58r2vnzc02aqrgcupzytm7nek6p8gwvfmg27uwjrp92q
0n77qtrsyu48ualq5q0vsjq0c8zssp3w30 md1f5nl58r3ujfumqccaemfyfg3cqu633yeh80klhqknfyd4kvjsnwn05ng0clf9h7z2z4ecw --chain 0 --cou
nt 5
bc1qr6h5gahcaqa8a35p3ts0d2w6qvhmsn7dhunu5xd9kyculcgz3dwqf266zj
bc1qr6357waaru5m82sde0xta75q6nj29gsd3m5krvxllekq89fs53lqwxkwza
bc1q3774tl4ntpwh0t842cdxtw2ljy82s0c47see7lujg5hs0zcqjk8s6tjlw4
bc1qshj4vra0cdcautnksujvh327dtxn394e2npvx6lcrdc4vrrhcv5qf6ga0u
bc1qyld70fxhz9x3mn2h5d58yal2hcgpn2na0a8d8pzsh0qd6lucnw2qs3serj
note: stdout is watch-only — public keys only, cannot spend
[exit 0]

$ /scratch/code/shibboleth/descriptor-mnemonic/target/debug/md address md1f5nl58qpp2040veppfx8qqxpx2v6aahst9z5qtlk2pxnhcj8c9
w4uzfyu0x4ggr0ff7jqeh2qcmjtmnm68 md1f5nl58qdgt06453fusyccggzn9xdwqqqyqqpxdweecy63hcar79u3zfygx57kvxs5x9uqf93jqxfl9ppns md1f5
nl58qhmwqmkran74xxmyryu7mk45vzzuv5e4kqq3aqgz4fntvqp28rqw5ws7hc37lyqg66puucsn7wgk md1f5nl58qm5578mfalmzaps9gsnxvyuyxxlwa7acux
mfpgptf2sxxjqu7cxmw8gzhnklf0sz7u90d6z6x4vj md1f5nl58pxh36ehykyezegzrvq3ltnhcyfxpvcss78u59v8gsu55wj3crtt4vs33fy3jupswd3pht0yq
kvmr md1f5nl58pd6uq0h2z2mwrxjfvd9tg0crh8n4t07049utsul4w5dyeanq0c5j5e0v2yeazrswfymhg943ez9q md1f5nl58ph3xdp7jdwd4x23agzgmtyet
a0sgkvqq005enx2qvg99s9s4zypw88eyeyvszvszk2mu8es7jcj3 md1f5nl58plxfygzzdkmhugqfedcfejakelta7k97h5hapszwrfazruxtax3rqls29j4mj2
q94ztc62lf0f9p md1f5nl58zpjrhst0rkv9m9e77su2nf9sgxcdxq2lqewdp0jr30af03vhes8kuf8wsfhhs7qxv0acr9njq92n md1f5nl58zd24tpdkh3prfu
7f47g4ntt488aal33v7xgtm7nzyxeyt99rm5cmeavkayw4e0s24a9vcq6xq8de md1f5nl58z4nmhnfk2h56yhptk4lf27gvugwccqqaqx5vfn7xcy7t7l04mrqu
hq5k7vj0ywqmrddyhf20urlg md1f5nl58z6t25tltp3ckcyejn3mjr868cj73pd0pw3euqyg9glwyfhcsvvsp9u8hdqmfh3syex2yyjd4kn6f md1f5nl58rxpc
8xhhu3sqxhwsyzw9jr32lcjrputwmdtlut5elkhtynnq7tm8drts47t2qkqmx3ulr0fwgpkm md1f5nl58r2vnzc02aqrgcupzytm7nek6p8gwvfmg27uwjrp92q
0n77qtrsyu48ualq5q0vsjq0c8zssp3w30 md1f5nl58r3ujfumqccaemfyfg3cqu633yeh80klhqknfyd4kvjsnwn05ng0clf9h7z2z4ecw --chain 1 --cou
nt 5
bc1q70d7dmaz2s4ur98vewnzyuuct6wzu3jpwhhhn8dz0g5agkevzjaqnn6ecs
bc1qgw504j2sphtt3rpy4hthwgnlgm3fyv8h7pc8nzeq5k68z6h75kzs35gn79
bc1qm7dxt9ghlf3nh5ump80vhyl9l23txpp5qqjcun2s29xcgfqjwndqgcwr42
bc1qsgqp0dwmatl72kdkj3gnhwkqrq9seu3pfpshlwtmnam82ktdp37qflqt8p
bc1qdmtv44gaj5e77apkzk9gfrhntrtn5nmn86qts8wuw7mnmtkwxkusvve7kd
note: stdout is watch-only — public keys only, cannot spend
[exit 0]
```

The key cards bind to the TEMPLATE id

An mk1 card carries a `policy_id_stub`, and the stub is **form-aware**: a card minted against the keyed policy roots on the wallet-policy id, while one minted against the keyless template roots on the template id. Same wallet, two ids — so a card made for the wrong form is refused at membership before seating is even attempted.

```
$ cat /scratch/code/shibboleth/mnemonic-engrave/design/journeys/out/rcw/tr/keys.txt
[39ec1b6e/270028'/0'/8'/0']xpub6Dn2FdZCkR6b6nCFWKMZraCwEcC7aedkRrVHf8DK5gSBrny6YJkiUZp9bL8j2a4osHSuhDsk1FBnSS81C82qvZ4uiDofJ
GQKmNmmxqfUxCr
[d02bcedf/270028'/0'/8'/0']xpub6DdqV8W2dzVkyfWmNUJHkoWdH6JctbyGvKrJVUDUL1hSoUBGtfB3JtXoyYtvkVR3S6jyzagALehL56vqwcjXAc5hoaCoL
6yyrtCLQf1dg2f
[5fd78eb3/270028'/0'/8'/0']xpub6FMyX6RDPGTJgt3hYThGymmhRUhntuA9HagmHmDPUm11hDKk6HLNFspSyRWqqdiSQYMQujjsL2Xn9ERJE9YDZR9jG2X4U
Jbx53XE3b8Z5Bw
[92c4c8b2/270028'/0'/8'/0']xpub6E9jU3J1VtuNb5GVBatKMhYkHiMM7dD6yK2x4fhDDSghXHQLxJQgFZnwLEXWc1jAb5ZCsNUhQGDqUdy3Jt3kRmSMG6Fny
BnM1PRXmD6bXb4
[086c047e/270028'/0'/8'/0']xpub6Ep7XhWjqPzjf5HTCQTf8M8QRYUqGHJLpXQ9qbU7xJrGGCsG9Fba4QUtughDTBLJgf5praqaxKYUhmTvKMsWiG8Gs24jJ
gLzspG2tstkBdz
[cef8224c/270028'/0'/8'/0']xpub6EvqkLWC3QnMavFkcHB5Kzfnv8FeikdNwurio9zyTWgNnwdsZXf7KdcGAh9to36zwLckiVkLpyjBUxD9f9xWSuT9KUmw1
v1QqzqDuYuBW7q
[exit 0]

note: stdout is watch-only — public keys only, cannot spend
$ /scratch/code/shibboleth/mnemonic-engrave/design/journeys/mk_cards_from_json.py /scratch/code/shibboleth/mnemonic-engrave/
design/journeys/out/rcw/tr/mk-encode.json /scratch/code/shibboleth/mnemonic-engrave/design/journeys/out/rcw/tr/mk-encode-ra
w.txt
6 key cards -> 18 mk1 chunks
[exit 0]

template-id prefix: 68a1a888   stub the cards embed: 68a1a888
```

The transcript gates on that equality rather than printing it: if the cards ever bound to the policy id instead, the run goes red.

Two wallets, and the proof they are two

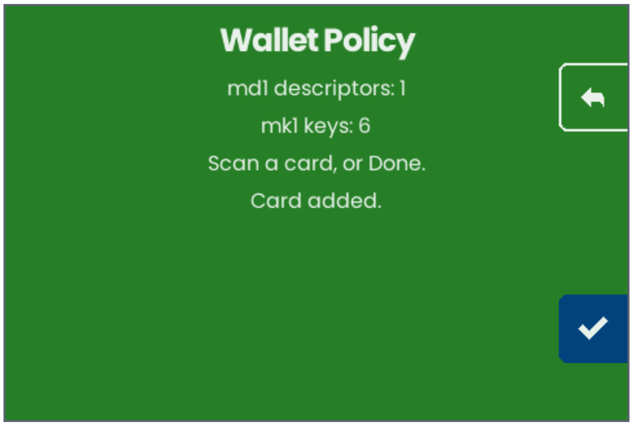
```
  wrapper  template-id                       first receive address
  tr       68a1a888385797337ce5debc90fcfb1e  bc1p8h77a4mgl49mld6crpdpvx4jsg6gclkv9vr93enl54cdz0gp6v8smuqr2n
  wsh      daee67be4eacf85e8b832ae64fc06566  bc1qr6h5gahcaqa8a35p3ts0d2w6qvhmsn7dhunu5xd9kyculcgz3dwqf266zj

Different ids, different addresses, as they must be.
```

This is gated, not observed. If a future edit ever made the two wrappings produce the same template id or the same addresses, the transcript fails — because at that point one of them is not the wallet the document says it is.

The device half — four arms

The same cards carried into the SeedHammer II emulator over NFC, by `capture_rcw.py`. Two routes per wrapper, and the second is the one that matters:

- **keyed** — the 15-chunk card alone. It carries its own keys, so nothing is seated. Proves the device can derive this wallet at all.
- **seating** — the 5-chunk keyless template plus its six mk1 cards, which is *what actually gets engraved*. Proves the device can join them.



tr / seating: 5 template chunks assembled, 6 key cards gathered.

tr / seating: the addresses, seated from the key cards.

tr — both routes agree

on the device AND on the host
`bc1p8h77a4mgl49mld6crpdpvx4jsg6gclkv9vr93enl54cdz0gp6v8smuqr2n`
`bc1panm4lv4hl8n96xccufjc7rkd4hhuvled2hp5eqdhncaqvpuvvfdq5z9t8t`
`bc1pt7g96plv4mw73vrs64wvq2eswq9zrh4r0ztnx7huuma94xzskkhqm8vkn4`
`bc1pp3fpcxm9zkp75hguhj5qh804wxzrff5tp6wcunr0up7j3xp46v3su0mntq`

Keyed and seating produced **the same** addresses.

wsh — both routes agree

on the device AND on the host
`bc1qr6h5gahcaqa8a35p3ts0d2w6qvhmsn7dhunu5xd9kyculcgz3dwqf266zj`
`bc1qr6357waaru5m82sde0xta75q6nj29gsd3m5krvxllekq89fs53lqwxkwza`
`bc1q70d7dmaz2s4ur98vewnzyuuct6wzu3jpwhhhn8dz0g5agkevzjaqnn6ecs`
`bc1qgw504j2sphtt3rpy4hthwgnlgm3fyv8h7pc8nzeq5k68z6h75kzs35gn79`

Keyed and seating produced **the same** addresses.

> **That equality is the whole point.**
> One wallet reached two ways — from a card that carries its keys, and from a template plus six separate key cards — across the air gap, through two implementations in two languages. Either route alone would only prove the device is self-consistent.

Five on the host, two on the device

The host derives five addresses per chain. The device shows **two**, and that is deliberate: `addrProofPerChain = 2` (`gui/wallet_policy.go:184`), chosen by plan D6 as *"receive AND change, FEWER of each — change is where a policy mismatch silently loses funds, so proving both chains derive beats proving one chain five times."*

Asked to reconsider for this journey, the ruling was to leave it (`design/agent-reports/RULING_sh2_address_count.md`). The argument: **every realistic mismatch diverges at index 0**, and an index-conditioned backdoor defeats a five-address screen exactly as easily as a two-address one. The extra three addresses buy reading fatigue on a *consent* surface and no additional evidence.

> So the device proves a
> **sample**
>
> and the host proves the **range**, and the walk asserts the device's four are byte-equal to the host's indices 0–1 per chain — rather than quietly comparing fewer and calling it agreement.

The comparison can fail

An assertion that addresses *appear* would pass for a walk that silently stopped driving the device. `--prove-it-can-fail` corrupts one character of one expected address and requires the walk to refuse; it exits 0 only when the walk fails.

Getting it to the device: the payload route

```
`me sysw pack` builds the SYSTEMWIDE container the device reads. This journey's
wallet carries no ms1 secret share -- the three preimages are engraved as text,
not as ms1 -- so the payload holds only public strings.

Strings for the payload: 46 total, ms1 count = 0

$ /scratch/code/shibboleth/mnemonic-engrave/target/debug/me sysw pack --in /scratch/code/shibboleth/mnemonic-engrave/design/
journeys/out/rcw/payload-strings.txt --out /scratch/code/shibboleth/mnemonic-engrave/design/journeys/out/rcw/payload.bin --n
o-passphrase
strength: no passphrase — BELOW the threshold
digest:   be81 c617 9403 3c05 0287 480a af95 5d44
runcap: CAPTURE FAILED -- no stdout line matched /./
[exit 1]

Payload: 3887 bytes.

$ /scratch/code/shibboleth/mnemonic-engrave/target/debug/me sysw show /scratch/code/shibboleth/mnemonic-engrave/design/journ
eys/out/rcw/payload.bin
sealed:   false
pub_len:  3835
ct_len:   0
identity: 04ca5f8938a7df7897af8a169ad9bc0595dfeaa93815b9a5b9cce402eff7f06f
digest:   be81 c617 9403 3c05 0287 480a af95 5d44
public record 0: md1/mk1 — confirmed
public record 1: md1/mk1 — confirmed
… 41 lines elided …
public record 43: md1/mk1 — confirmed
public record 44: md1/mk1 — confirmed
public record 45: md1/mk1 — confirmed
[exit 0]

public records in the container: 46   strings fed in: 46
… clipped; full text in transcript_rcw.txt
```

Gated both ways: the container's record count must equal the number of strings fed in, and the ms1 count must be **zero**
— this wallet's three passphrases are engraved as text, not as ms1 shares.

The sealed route, and the limit it has

```
`me seal` encrypts the records and emits a UF2 for `picotool load`. The
passphrase is GENERATED, never supplied -- a memorable one does not survive an
offline attack on a stolen machine.

IT IS SEALED PER WRAPPER, AND NOT BY CHOICE. `me seal` accepts 1..=24 records;
the two bundles together are 46, so the combined set is REFUSED. Each wrapper's
23 strings fit with one to spare. That is a real limit of this route and the
unsealed `sysw pack` above does not share it -- it took all 46.

$ /scratch/code/shibboleth/mnemonic-engrave/target/debug/me seal --in /scratch/code/shibboleth/mnemonic-engrave/design/journ
eys/out/rcw/payload-strings.txt --out /scratch/code/shibboleth/mnemonic-engrave/design/journeys/out/rcw/sealed-both.uf2 --it
erations 100000
me: 46 records; must be 1..=24 (2-of-3 multisig is 15)
[exit 4]

Refused, as documented. Now per wrapper:

-- tr: 23 records
$ /scratch/code/shibboleth/mnemonic-engrave/target/debug/me seal --in /scratch/code/shibboleth/mnemonic-engrave/design/journ
eys/out/rcw/tr/backup-strings.txt --out /scratch/code/shibboleth/mnemonic-engrave/design/journeys/out/rcw/tr/sealed.uf2 --it
erations 100000
me: wrote 4096 bytes to /scratch/code/shibboleth/mnemonic-engrave/design/journeys/out/rcw/tr/sealed.uf2

passphrase — write this down and store it APART from the machine:

    wave immense clay speak hour better lyrics fringe stock section outdoor radar

load:  picotool load --verify /scratch/code/shibboleth/mnemonic-engrave/design/journeys/out/rcw/tr/sealed.uf2   (machine in
BOOTSEL)
wipe:  picotool erase -r 0x10E00000 0x10E10000
runcap: CAPTURE FAILED -- no stdout line matched /./
[exit 1]

   sealed UF2: 4096 bytes

-- wsh: 23 records
$ /scratch/code/shibboleth/mnemonic-engrave/target/debug/me seal --in /scratch/code/shibboleth/mnemonic-engrave/design/journ
eys/out/rcw/wsh/backup-strings.txt --out /scratch/code/shibboleth/mnemonic-engrave/design/journeys/out/rcw/wsh/sealed.uf2 --
iterations 100000
me: wrote 4096 bytes to /scratch/code/shibboleth/mnemonic-engrave/design/journeys/out/rcw/wsh/sealed.uf2

passphrase — write this down and store it APART from the machine:

    chronic play multiply sight various celery awesome guess noble sense cloud mother

load:  picotool load --verify /scratch/code/shibboleth/mnemonic-engrave/design/journeys/out/rcw/wsh/sealed.uf2   (machine in
BOOTSEL)
wipe:  picotool erase -r 0x10E00000 0x10E10000
runcap: CAPTURE FAILED -- no stdout line matched /./
[exit 1]

   sealed UF2: 4096 bytes
```

The two routes do not have the same capacity.

`me seal` accepts 1..=24 records; both bundles together are 46, so the combined set is **refused**. Each wrapper's 23 fit, with one to spare. The unsealed `sysw pack` took all 46 without complaint.

The transcript demonstrates that refusal and then gates on it still happening. If the cap is ever raised, that gate goes red and this page gets rewritten — rather than quietly becoming false.

Wallet files: what this wallet can and cannot export

```
`mnemonic export-wallet` (mnemonic-toolkit, shipped since v0.97.0) emits 11
formats. Run against BOTH concrete descriptors, every format, nothing hidden:

  wrap  format         result       detail
  tr    descriptor     EMITTED      1299 bytes
  tr    bsms           refused(2)  error: --format bsms does not support taproot (P2trMulti);
  tr    bip388         refused(1)  error: --format bip388 requires every descriptor key to en
  tr    bitcoin-core   EMITTED      2806 bytes
  tr    sparrow        refused(1)  error: --format sparrow requires --template; descriptor pa
  tr    electrum       refused(1)  error: --format electrum requires --template; descriptor p
  tr    specter        refused(2)  Re-invoke with all missing fields supplied.
  tr    jade           refused(1)  error: --format jade requires --template (wsh-multi / wsh-
  tr    coldcard       refused(1)  error: --format coldcard requires --template (bip44 / bip4
  wsh   descriptor     EMITTED      1243 bytes
  wsh   bsms           EMITTED      1325 bytes
  wsh   bip388         EMITTED      1350 bytes
  wsh   bitcoin-core   EMITTED      2694 bytes
  wsh   sparrow        refused(1)  error: --format sparrow requires --template; descriptor pa
  wsh   electrum       refused(2)  error: --format electrum cannot faithfully export an UNSOR
  wsh   specter        refused(2)  Re-invoke with all missing fields supplied.
  wsh   jade           refused(2)  error: --format jade cannot faithfully export an UNSORTED
  wsh   coldcard       refused(2)  error: --format coldcard cannot faithfully export an UNSOR

6 of 18 (wrapper x format) combinations emitted a file.

THE TAPROOT FORM EMITS NOTHING, and every format fails at the SAME place --
… clipped; full text in transcript_rcw.txt
```

The taproot form emits nothing at all

, and every format fails at the same place — before format selection is even reached. That is `rust-miniscript`'s `sanity_check()` refusing tier 4, which needs no key.

Why no wallet app will take it — one reason, three apps

Measured, not assumed. Against a real Bitcoin Core node, isolating one feature at a time:

probe	Core
`wsh(and_v(v:sha256(H1),pk(K)))` — hashlock AND a key	accepts
`wsh(and_v(v:sha256(H1),multi(3,…)))` — our tier 1	accepts
`wsh(and_v(v:after(…),sha256(H3)))` — **tier 4 as-is**	**refuses**
the same, **with a key added to tier 4**	**accepts**
`wsh(or_i(keyed, KEYLESS))`	**refuses**
`wsh(or_i(keyed, keyed))`	accepts

Not the hashlocks. Not the mixed timelock flavours. Not `multi_a`, `or_i`, or the NUMS key. The single sigless spend path, and nothing else.

All three named apps refuse for that reason. Nunchuk reaches it through libnunchuk's `IsSane()`/`NeedsSignature()` — the same vendored Core checker. Sparrow refuses by a different mechanism: it has no miniscript engine at all.

> **In Core the rule is non-waivable.**
> No version through v31.1 loads this wallet, either wrapper, watch or hot. Relaxing our own gate would produce a file no target accepts.

> Version floors, pinned on six real binaries: `wsh` miniscript watch v24 / sign v25, `tr` miniscript v26, and multipath `<0;1>` only v29+ — absent even from v28, and undocumented in the release notes that introduced it. The one Core-loadable watch artifact for this wallet is an `addr()` list: you can watch it, you cannot describe it.

Three implementations, one address

The BSMS record's fourth line is BIP-129's canary — the wallet's first address, computed by the *emitter*. Set against `md`'s own derivation, and against what SeedHammer II showed after seating six key cards onto the keyless template over NFC:

```
  BSMS canary (mnemonic-toolkit emitter) : bc1qr6h5gahcaqa8a35p3ts0d2w6qvhmsn7dhunu5xd9kyculcgz3dwqf266zj
  md address  (descriptor-mnemonic)      : bc1qr6h5gahcaqa8a35p3ts0d2w6qvhmsn7dhunu5xd9kyculcgz3dwqf266zj
  SeedHammer II (Go, over NFC, seated)   : bc1qr6h5gahcaqa8a35p3ts0d2w6qvhmsn7dhunu5xd9kyculcgz3dwqf266zj

ALL THREE AGREE — three implementations, two languages, one air gap.
```

> Three implementations, two languages, one air gap, and a BIP-129 record produced by a fourth code path — all naming the same address. The transcript fails if they ever stop.

What this journey does NOT show

- **No real hardware.** The device half is the emulator — the same Go firmware compiled to wasm, driven over the same NFC entry point, but not a machine with a screen and a hammer.
- **Nothing was engraved.** The plates are rendered previews. No steel was cut, and the payloads were built but not flashed to a device.
- **No wallet app actually loaded these files.** The four wsh artifacts were emitted and their BSMS canary cross-checked, but no Nunchuk, Sparrow or Core instance was driven to import one — and per the reports, none of them would. The Core refusals were measured directly; the Nunchuk and Sparrow verdicts rest on source reading plus Core execution, not on a live import.
- **No hot wallet.** Every artifact here is watch-only. Hot export exists nowhere in the constellation and is deliberately gated on an explicit go-ahead — it writes spendable key material to disk.
- **No spend.** Every address here is derived, none is funded, and no tier has been exercised against a real chain. The timelocks in particular are arithmetic in this document and consensus rules everywhere else.